

Minor-1CSL7640 - Natural Language UnderstandingNOTE:

1. This is Question-cum-Answer sheet. Maximum Points: 50, Total Questions: 15, Total Page: 10 (including both side), Total Time: 2 Hour. **If there is anything not clear in the problems, go ahead with your own assumptions but state them clearly. No doubts will be entertained during the exam.** Be precise and write the answer in the box provided. Verbosity will be penalized. Use the other answer sheet for rough work and submit both.

• Name:

Roll Number:

Signature:

Question Number	Topic	Total Marks	Marks Obtained
1-10	Short Answers	20	
11	Backprop	5	
12	HMM	5	
13	Edit Distance	5	
14	Latent Semantic Analysis	5	
15	Word2Vec	10	
	Total	50	

Short Answer Questions Q1-Q10: 2 points each

1. What does the following Regular Expression mean: $\sim[01]^*1\$$.

Binary string that ends with one and start with ~~one~~ zero or more 01.

2. We have learned that dense word vectors learned through Word2Vec or GloVe have many advantages over using sparse one-hot word vectors. Which of the following is **NOT** an advantage that dense vectors have over sparse vectors?

- ☒ A. Models using dense word vectors generalize better to unseen words than those using sparse vectors.
☒ B. Models using dense word vectors generalize better to rare words than those using sparse vectors.
☒ C. Dense word vectors encode similarity between words while sparse vectors do not.
☒ D. Dense word vectors are easier to include as features in machine learning systems than sparse vectors.

A

3. If your training loss increases with the number of epochs, which of the following could be a possible issue with the learning process?

- A. Regularization is too low and the model is overfitting.
 B. Regularization is too high and the model is underfitting.
☒ C. Step size is too large.
 D. Step size is too small.

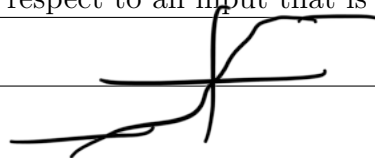
C / C, B can be considered but not B alone

4. Assuming characters are properly segmented, if one models OCR as Hidden Markov Model what will be observed state and what will be hidden state?

$[A][B][C] \rightarrow ABC$ H: Characters
 O: features of Char B B

5. What would the gradient of the sigmoid be with respect to an input that is very large?

tends to zero



6. Give one example of gender bias in word embeddings.

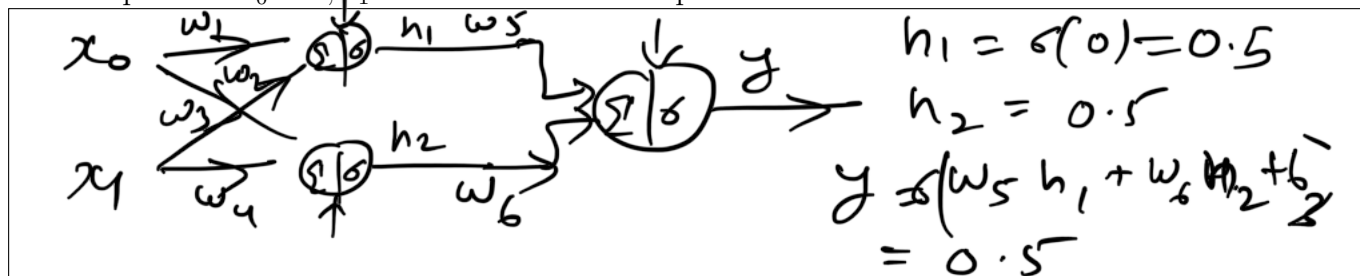
man: doctor ; woman: Nurse

7. Consider the handwritten digit-classification problem trained using cross-entropy loss. Assume the model outputs a uniform probability distribution over all the classes for every input. What is the resulting cross-entropy loss?

$$C.E. = -\log \frac{1}{e} = \log_2 10 \approx 3.32 \checkmark$$

$$\log_e 10 = 2.3 \checkmark$$

8. Consider modeling X-OR problem using a simple neural network. Take Square Error as divergence function. Use following network: 2 inputs (x_0, x_1), 2 hidden neurons, sigmoid activation, 1 output neuron, sigmoid output. Initialize all weights and bias to zero. Draw the network and do forward pass for $x_0 = 1, x_1 = 1$ and show the output of each neuron.



9. Consider the sentence: "I love IIT Jodhpur." Using the skip-gram model where the window size is defined as the maximum distance from the center word, and the window size is 3, list all the positive (center, context) word pairs generated from this sentence.

(I, love), (I, IIT), (I, Jodhpur)
 (love, I), (love, IIT), (love, Jodhpur)
 IIT
 Jodhpur

12 pairs

10. True or False: For representing rare words, the CBOW variant of Word2Vec performs better than the Skip-Gram variant. Justify your answer with a brief explanation.

False

Backpropagation

11. Consider the computational graph for the following expression $f = \frac{1}{1+e^{-(w_0 x_0 + w_1 x_1 + w_2)}}$ shown in the next page. The initial values for variables w_0, w_1, x_0, x_1 and w_2 are also shown in the graph. (a) Using these initial values perform, Forward Pass and compute the numerical value at every intermediate edge in the graph using the given initial values. Write the forward-pass numerical value above each edge in the computational graph. (1 point) (b) Using backpropagation, compute the gradients of the final output f with respect to: w_0, x_0, w_1, x_1, w_2 . Write the backward-pass gradient (the gradient flowing through that edge) below each edge in the computational graph. (4 points)

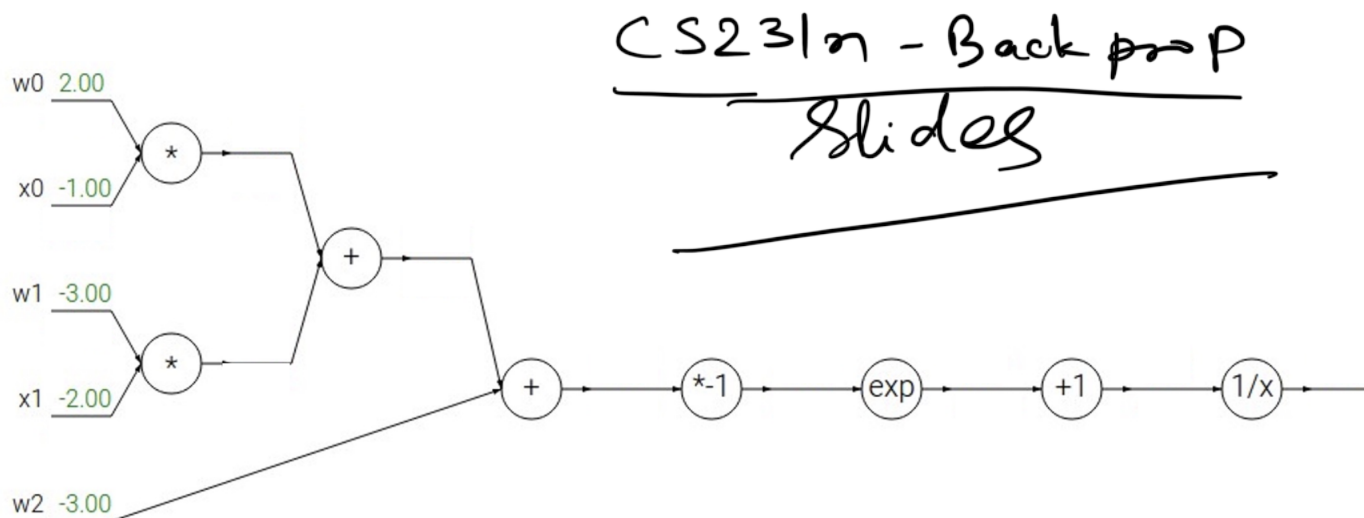


Figure 1:

Hidden Markov Model

12. Consider a Hidden Markov Model (HMM) for Part-of-Speech (POS) tagging with two tags: Noun (NN) and Verb (VB). The initial, transitional and emission probabilities are given below (here $\langle s \rangle$ is the start symbol):

Initial probabilities:

$$P(NN | \langle s \rangle) = 0.6, \quad P(VB | \langle s \rangle) = 0.4$$

Transition probabilities:

$$P(NN | NN) = 0.7, \quad P(VB | NN) = 0.3$$

$$P(NN | VB) = 0.4, \quad P(VB | VB) = 0.6$$

Emission probabilities:

$$P(dog | NN) = 0.5, \quad P(dog | VB) = 0.1$$

$$P(barks | NN) = 0.1, \quad P(barks | VB) = 0.6$$

- (a) Given the observation sequence: dog barks. Using the Viterbi algorithm: determine the most probable sequence of POS tags and its probability. Show all steps. (4 Points)

$$\begin{aligned}
 \langle s \rangle \langle NN \rangle &= P(NN | \langle s \rangle) \times P(dog | NN) = 0.3 \\
 \langle s \rangle \langle VB \rangle &= 0.4 \times 0.1 = 0.04 \\
 \langle s \rangle \langle NN \rangle \langle VB \rangle &= 0.3 \times 0.3 \times 0.6 = 0.054 \\
 \langle s \rangle \langle NN \rangle \langle NN \rangle \\
 \langle s \rangle \langle VB \rangle \langle NN \rangle \\
 \langle s \rangle \langle VB \rangle \langle VB \rangle
 \end{aligned}$$

(b) For a real world setup, what data is required to learn the Initial, Transition, Emission probabilities. (1 Point)

(Sentence, pos tag)

Edit Distance

13. Compute the **Edit Distance** between the two strings: horse and hotel. Assume the following operation costs: Insertion: cost = 1, Deletion: cost = 1, Substitution: cost = 2. Now,

(a) Define the dynamic programming recurrence relation. (1 Point)

$$D[i][j] = \begin{cases} D[i-1][j] + 1 & \text{del} \\ D[i][j-1] + 1 & \text{insertion} \\ D[i-1][j-1] + c & \text{sub.} \end{cases}$$

$$c = \begin{cases} 0 & \text{if } x_i = y_j \\ 2 & \text{if } x_i \neq y_j \end{cases}$$

(b) Clearly specify the base cases. (0.5 Points)

$$\begin{aligned} D[0][0] &= 0 \\ D[i][0] &= i \\ D[0][j] &= j \end{aligned}$$

(c) Construct the complete DP table and mark the final edit distance. (2.5 Points)

	e	h	o	t	e	l
e	0	1	2	3	4	5
h	1	0	1	2	3	4
o	2	1	0	1	2	3
r	3	2	1	2	3	4
s	4	3	2	3	4	5
e	5	4	3	4	3	4

(d) Provide one optimal sequence of edits to convert horse into hotel. (1 Point)

Insert l → Match e → Sub. s → t → Del r
→ match o → match h.

Latent Semantic Analysis

14. Consider the following corpus with three documents: D1: cat dog; D2: cat tiger; D3: car bus. Answer the following questions:

(a) Construct the vocabulary of the corpus.

(0.5 point)

$V = \{cat, dog, tiger, car, bus\}$

(b) Construct the term-document matrix A using term frequency. Clearly specify the rows and columns.

(0.5 point)

	D1	D2	D3
cat	1	1	0
dog	1	0	0
tiger	0	1	0
car	0	0	1
bus	0	0	1

(c) Compute the matrix $A^T A$.

(1 point)

$$A^T A = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

(d) Find the eigenvalues of $A^T A$ and the corresponding singular values of A . (2 points)

$$\lambda = 1, 2, 3$$

$$\sqrt{3}, \sqrt{2}, \sqrt{1}$$

(e) Briefly explain what the largest singular value represents in the context of Latent Semantic Analysis (LSA). (1 point)

The strongest underlying semantic pattern.

Word2Vec

15. (a) Write the objective function J of the Skip-gram with Negative Sampling (SGNS) model for a single training instance consisting of: center word embedding v_c , one positive (context) word embedding u_0 , and K negative word embeddings u_k .

$$J = -\log \sigma(u_0^T v_c) - \sum_{k \in K} \log \sigma(-u_k^T v_c)$$

- ① (b) Compute the explicit gradients of the objective function with respect to: $\frac{\partial J}{\partial v_c}$, $\frac{\partial J}{\partial u_0}$, $\frac{\partial J}{\partial u_k}$. Show all intermediate steps in your derivation.

done in class

(c) Derive the gradient update rules for the embeddings v_c , u_0 , and u_k under stochastic gradient ascent (or descent). Clearly indicate the direction of update. (3 points)

$$u_0 = u_0 - \eta \frac{\partial J}{\partial u_0} \dots$$

(d) The following implementation of Skip-gram with Negative Sampling (SGNS) training segment contains **three conceptual mistakes**. Identify and correct them. (6 points)

```

1 learning_rate = 0.025
2 epochs = 1000
3 num_negative = 5
4
5 for epoch in range(epochs):
6     total_loss = 0
7     random.seed(42)
8     random.shuffle(training_pairs)
9
10    for center_idx, context_idx in training_pairs:
11        v_c = W_center[center_idx]
12        grad_v_c = np.zeros(embedding_dim)
13
14        # Positive sample
15        u_o = W_context[context_idx]
16        z_pos = np.dot(v_c, u_o)
17        p_pos = sigmoid(z_pos)
18
19        loss_pos = -np.log(p_pos + 1e-9)
20        total_loss += loss_pos
21
22        err_pos = p_pos
23
24        grad_v_c += err_pos * u_o
25        W_context[context_idx] -= learning_rate * err_pos * v_c
26
27        # Negative samples
28        neg_indices = sample_negatives(num_negative, context_idx)
29
30        for neg_idx in neg_indices:
31            u_k = W_context[neg_idx]
32            z_neg = np.dot(v_c, u_k)
33            p_neg = sigmoid(z_neg)
34
35            loss_neg = -np.log(p_neg + 1e-9)
36            total_loss += loss_neg
37
38            err_neg = p_neg

```

$err_{pos} = p_{pos} - 1$
 $loss_{neg} = -np.log(1 - p_{neg} + 1e-9)$

```
39         grad_v_c += err_neg * u_k
40         W_context[neg_idx] -= learning_rate * err_neg * v_c
41
42     W_center[center_idx] -= learning_rate * grad_v_c
```

Assignment- 2 — ① Word2vec

② RNN

March 20

Paper Reading: - 1